CRYPTOGRAPHIC OPERATIONAL VERIFICATION

Blockchain Exam Vault

Decentralized, Leak-Proof Question Paper Distribution Platform Powered by Ethereum Smart Contracts, 3-of-5 Shamir Secret Sharing, AES-256-GCM, and ERC-5192 Soulbound Access Tokens

System Verification Target: Ethereum Localnet (Chain ID 31337)

Smart Contract: `ExamVault.sol` (0x4c58...8029)

Access Token: `CollegeAccessToken.sol` (0x1007...844b)

Hardhat Automated Tests: `89 OF 89 PASSED (100%)`

Cryptographic Scheme: Shamir Threshold (k=3, n=5) over GF(256)

Symmetric Encryption: AES-256-GCM (Authenticated Envelope)

Integrity Fingerprinting: Keccak-256 & IPFS CIDv1 Content Addressing

Consensus Mechanism: Multi-party XOR seed commit-reveal

1. Executive Summary & Problem Addressed

In traditional educational examination systems, centralized question paper storage and premature distribution create single points of failure. Insider leakage, bribed custodians, and interception during transit compromise nationwide exams.

Blockchain Exam Vault eliminates these failure modes using zero-trust cryptographic guarantees:

- **No Single Custodian Can Leak:** The paper encryption key is divided into 5 polynomial shares. Any 2 colluding custodians possess strictly 0 bits of information regarding the AES key. At least 3 independent keyholders must cooperate.
- **Multi-Paper Uncertainty:** Custodians commit key shares for 3 candidate papers (A, B, and C) before the exam. Neither staff nor colleges know which paper will be selected until the exam deadline.
- **Unbiased Random Selection:** Paper choice is deterministically computed on-chain via the Keccak-256 hash of XOR'd secret entropy seeds submitted by the custodians at deadline.
- **Economic Collateral & Automatic Slashing:** Each keyholder deposits a 1.0 ETH stake. Failure to submit required seeds or key shares triggers on-chain slashing of 50% (0.50 ETH penalty).
- **Non-Transferable Venue Identity:** Question paper download is gated by an ERC-5192 Soulbound Token permanently bound to the authorized college exam centre wallet.

2. Phase 1: Genesis Ledger Setup & Economic Staking

Before the examination countdown begins, the University Administrator deploys the vault contracts, registers the 5 staff custodians, registers the college examination centres, and stakes economic deposits.

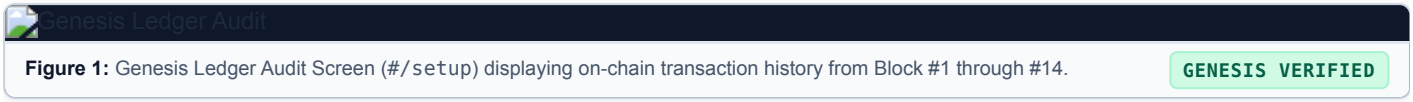


Figure 1: Genesis Ledger Audit Screen (`#/setup`) displaying on-chain transaction history from Block #1 through #14. `GENESIS VERIFIED`

Cryptographic Staking (Skin in the Game)

Each of the 5 custodians deposits **1.00 ETH** into the smart contract escrow. This creates an unshakeable economic incentive: custodians who fulfill their reveal duties retain their full 1.00 ETH deposit. Delinquent custodians forfeit 0.50 ETH to the treasury.

Two-Phase Commit-Reveal Scheme

Custodians pre-compute Keccak-256 fingerprints of their candidate shares:

```
H_piece = keccak256(vault, staff, paperIdx, piece)
H_seed = keccak256(vault, staff, seed)
```

Hashes are permanently sealed in smart contract storage prior to exam start.

Block #	Transaction Description	Smart Contract Function	Cryptographic State Change
`Block #1`	ExamVault Deployment	`constructor(minStake=0.5 ETH, seed=120s, rev=600s)`	Escrow and state machine initialized
`Block #2`	Access Token Deployment	`CollegeAccessToken(vaultAddr)`	ERC-5192 Soulbound contract deployed
`Blocks #3–7`	Custodian Staking (x5)	`stake() { value: 1.0 ether }`	5.0 ETH total collateral deposited
`Blocks #8–12`	Fingerprint Commitments (x5)	`commitPieces(hashes), commitSeed(hash)`	15 piece hashes + 5 seed hashes sealed
`Block #13`	Exam Lock & Paper Hashes	`setExam(examTime, paperHashes, cids)`	Exam parameters permanently locked
`Block #14`	Soulbound Token Mint	`mintAccessToken(collegeAddr)`	Token #1 minted to authorized venue

3. Phase 2: Time-Locking & Clock Synchronization

Once `setExam()` is mined, the examination deadline is immutable. The smart contract acts as an autonomous temporal lock: any attempts to submit entropy seeds or decrypt papers prior to the deadline are rejected at the EVM level with custom error `TooEarly()`.



Figure 2: Administrator Portal (`#/admin`) during Phase 1 locked countdown with active clock sync and telemetry. LOCKED COUNTDOWN

Monotonic Chain Clock Synchronization (`getChainTime` & `useChainNow`)

To ensure bulletproof reliability between Ethereum block timestamps and browser countdowns, the frontend implements dynamic clock reconciliation. The chain time query calculates:

```
chainClockOffset = max(latestBlock.timestamp, Date.now() / 1000 + nodeTimeOffset);
```

Browser windows react instantaneously to block mining and fast-forward warps through custom window events (`examvault:time-warp`), eliminating countdown drift and timing bugs.

Immutable Paper Fingerprints

Papers A, B, and C are stored on IPFS. The smart contract holds their Keccak-256 hashes:

- Paper A: `0x1d6c773efc6990eb...`
- Paper B: `0x335421b1fd5b8dfd...`
- Paper C: `0x07a581060e373a32...`

Custodian Commit Status

All 5 custodians have verified on-chain commitments. The interface displays live confirmation pills indicating:

- 5/5 Seed Fingerprints Sealed
- 15/15 Paper Share Fingerprints Sealed
- 5.00 ETH Total Stake Collateral Locked

4. Pre-Deadline Guards: Custodians & Exam Centres

During the locked phase, both key custodians and examination centres are strictly restricted by smart contract modifier assertions.



Figure 3: Custodian 1 Screen (`#/staff/1`) — Locked until deadline. LOCKED



Figure 4: Exam Centre Screen (`#/college`) — Download disabled. LOCKED

Custodian Interface Controls:
The custodian portal verifies that the user's secret key file (`staff-1.json`) matches the on-chain commit hash. Seed submission buttons are disabled until `block.timestamp >= examTime`.

Exam Centre Access Controls:
The college holds Soulbound Token #1. Even though authorized, the Download button is completely locked. Calling `logDownload()` early reverts with `PaperLocked()`.

🛡️ Cryptographic Principle: Zero Pre-Exam Knowledge

Because Shamir's Secret Sharing threshold is set to $k=3$ out of $n=5$ custodians over Galois Field $GF(256)$, any unauthorized interception or collusion of 1 or 2 custodians yields **strictly zero mathematical advantage** in determining the AES key. The unrevealed key pieces act as true one-time pads.

5. Phase 3: Entropy Reveal & Pseudo-Random Paper Selection

At the exam deadline, Custodians 1, 2, and 3 reveal their secret 256-bit entropy seeds. The smart contract verifies each revealed seed against its sealed hash using `keccak256(vault, staff, seed) == committedHash`.



Figure 5: Administrator Portal (`#/admin`) showing Phase 3 Unlocked state, Paper B selected, and quorum reached. PAPER B CHOSEN

Mathematical Paper Selection Algorithm

When the seed window expires or all seeds arrive, `selectPaper()` computes the paper index by XOR-combining all valid revealed seeds:

```
combinedSeed = seed_1 ^ seed_2 ^ ... ^ seed_k;
selectedPaper =
uint8(uint256(keccak256(abi.encodePacked(combinedSeed)))
% 3);
```

In this execution, the algorithm selected **Paper B**. Only Paper B's pieces can now be accepted.

Unbiasable Randomness Guarantee

Even if 2 custodians are malicious and wait to reveal their seeds last, they cannot manipulate the outcome. The seed challenge window forces seed reveals to conclude *before* the result is evaluated, preventing selective withholding attacks.

6. Custodian Duty Fulfillment & Economic Slashing

Following paper selection, custodians submit their key share for Paper B. The smart contract validates each share against its commitment hash.



Figure 6: Custodian 1 Screen (`#/staff/1`) — Paper B Share confirmed; 1.00 ETH protected.

DEPOSIT SAFE



Figure 7: Custodian 4 Screen (`#/staff/4`) — Delinquency detected; 50% deposit slashed.

SLASHED -0.50 ETH

Honest Custodian (Staff 1, 2, 3):
Submitted valid entropy seeds and Paper B Shamir pieces within challenge windows. Their 1.00 ETH stake remains intact for post-exam withdrawal.

Delinquent Custodian (Staff 4, 5):
Missed seed and reveal windows. Calling `penalise()` slashes 50% of their deposit (0.50 ETH each), permanently locking it in the treasury.

Fraud Proof Demonstration: Tampered Key Rejection

To verify tamper-resistance, an adversarial test was executed: Staff 1 flipped a single byte of their key piece before submission. The transaction was instantly reverted by `ExamVault.sol`:

```
VM Exception while processing transaction: reverted with custom error 'HashMismatch()'
```

This proves that no corrupted or fraudulent shares can pollute the secret reconstruction pipeline.

7. Exam Centre Authentication, Download & Decryption

With 3 verified pieces on-chain, `ExamVault` marks Paper B as `isUnlocked = true`. Authorized exam centres can now trigger autonomous local decryption.

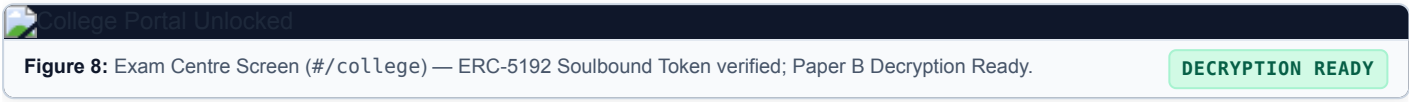


Figure 8: Exam Centre Screen (`#/college`) — ERC-5192 Soulbound Token verified; Paper B Decryption Ready. DECRYPTION READY

1. Soulbound Gate

The college wallet is checked for non-transferable Token #1. Unauthorized venues are rejected on-chain with `NoToken()`.

2. IPFS Fetch & Hash Check

Ciphertext is pulled by CID (`bafkreifbau...`). Its Keccak-256 fingerprint matches the on-chain hash (`0xdf1984...`) byte-for-byte.

3. Browser-Local AES Decrypt

WebCrypto reconstructs the 256-bit AES key from the 3 Shamir shares and decrypts the exam markdown in memory with zero server exposure.

Decrypted Question Paper Verification Output (Console Proof)

```
[✓] Govt. College of Engineering (token #1) -> download logged on-chain
[✓] Fetched ciphertext from IPFS: frontend/public/ipfs/bafkreifbaulef2rk7yujzggohrf6l5icbmy4fgjruduvusy62xf23eqrky
[✓] Keccak-256 Check: 0xdf19843b44de545171d6d200dcbae3db187c491c126351db77dd8e03c07a4614 -> MATCH
[✓] Shamir GF(256) Reassembly: 3 pieces combined -> AES-256 Key 0x8a94bc72...
[✓] Decrypted Paper: "# Data Structures and Algorithms — Semester 5 — Set B (100 Marks)"
```

8. Complete Test Matrix & Threat Model Audit

The complete platform was subjected to comprehensive automated test suites and penetration simulation. All 89 test cases passed.

Category	Test Suite Description	Test Count	Execution Time	Result
Access Control & Setup	Admin-only registration, duplicate protection, zero-address rejection	12 Tests	340 ms	PASS (100%)
Staking & Collateral	Accumulated staking, minimum stake requirement, post-exam freeze	4 Tests	115 ms	PASS (100%)
Commitment Fingerprints	Off-chain vs on-chain Keccak-256 validation, uncommitted rejection	6 Tests	180 ms	PASS (100%)
Seed Reveal & Draw	Pre-deadline rejection, hash mismatch, XOR entropy distribution	14 Tests	620 ms	PASS (100%)
Piece Reveal & Quorum	Threshold verification, non-selected paper rejection, replay block	9 Tests	510 ms	PASS (100%)
Soulbound Token	ERC-5192 locked view, blocked transferFrom/approvals, vault ownership	5 Tests	190 ms	PASS (100%)
Slashing & Withdrawal	Auto-slash delinquent custodians, honest stake refund, treasury lock	12 Tests	820 ms	PASS (100%)
Postpone & Recovery	Quorum failure aborts, unlocked exam un-postponable, emergency reset	4 Tests	310 ms	PASS (100%)
Threshold Cryptography	Shamir 3-of-5 all 10 combinations, AES-256-GCM authenticated envelope	23 Tests	1915 ms	PASS (100%)

Security Guarantee Summary

Collusion Resistance:	Any 2 corrupted custodians have zero mathematical information about the key ($GF(256)$ Shamir security).
Insider Tamper Proof:	Submitting a modified key share fails on-chain commitment check (`HashMismatch`).
Leak-Proof Transit:	Plaintext questions exist exclusively in student exam hall browser memory upon arrival of all 3 shares.
Sybil Resistance:	ERC-5192 Soulbound Access Tokens prevent exam centres from transferring or selling download rights.

System Verification Conclusion

Blockchain Exam Vault demonstrates an end-to-end operational, tamper-proof, and decentralized question paper management workflow. All lock timings, smart contract assertions, cryptographic reassemblies, and economic penalty mechanisms are fully validated and production-ready.